

DS4420 Final Project: Tennis Serve Speed Predictor

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1. Introduction

In sports, there are many metrics that can be observed, tracked, and eventually predicted. These can range anywhere from a player's physical characteristics such as height and weight, or how many points a player has scored across an entire season. In tennis, one such statistic is serve speed. The faster the serve, the harder it is to return and start a rally, resulting in aces and points for the player serving. Serve speed is a very variable metric, and it is dependent on the player's physical attributes along with body mechanics. For our final project, we developed a user-user collaborative filtering model and a bayesian linear regression model in order to attempt to predict this metric. By understanding which factors influence predictions the most, it could help players understand more about their playstyle and strengths, and possibly help with increasing their serve speed.

2. Literature Review

Previous research has explored the relationship between physical attributes and serve speed, which provided motivation for the predictors used in our models.

Vaverka and Cernosek (2013) conducted a study on elite tennis players in which they examined the relationship between height and serve speed, measuring their fastest serve, as well as average first and second serve speeds. The study suggests that there is a positive correlation between these metrics, with coefficients ranging from 0.48 to 0.63. The authors attribute this to a higher contact point, which allows for a steeper angle into the service box with greater speed. This study motivated us to include height as a predictor in our models.

Wong et al. (2014) conducted a study on the effects of BMI and full-body kinematics on serve speed. They found that body structure and mass have a significant correlation with serve speed. While their study emphasizes the importance of whole-body motion, including everything from the lower body to the arm, they suggest that physical attributes like BMI and weight are also meaningful predictors. This supports our inclusion of weight as a predictor in our models.

A study was also conducted to identify the relationship between age and skill level on tennis serve performance. It was found that skill level was more influential on improved serves compared to age, suggesting that elite players had higher performance due to their "greater hand, arm and hip peak velocities, as well as spatial differences in the hand location previous to ball impact" (Hernández-Davó et al. 2019). In our project, this research suggests that age is not a major factor in predicting serve speed but rather the skill level of the player is more important. Although we assume that older players would have more experience, there are many young players who play tennis at a high level.

Together, these studies support our modeling choices and suggest that while physical attributes are meaningful predictors of serve speed, they do not tell the full story.

3. ML Methodology

3.1 User-User Collaborative Filtering

In order to set up a user-user collaborative filtering model, we organized our data so that each row was a player, with columns containing metric data such as age, serve speed (km/h), height (cm), weight (kg), and dominant hand. We one-hot encoded the player's dominant hand since it was categorical. Then, the remaining numerical columns were min-max scaled while passing over any null values. Our CF model was able to calculate both cosine and L2 similarity scores, and we used k values (number of similar users) of 3, 5, and 10 to compare performances. In order to test the CF model, we randomly picked 10 athletes from the dataset and masked their serve speed. Then, we ran the model and calculated the RMSE and MAE to evaluate the performance. We also checked the relationship between RMSE and different k values in order to find an optimal k value to evaluate the model.

3.2 Bayesian Modeling

We fit a Bayesian linear regression model to predict professional tennis player serve speed (km/h) from three physical attributes: height (cm), weight (kg), and age. Prior to modeling, all three predictors were standardized. The model takes the form of $\text{speed} \sim N(\mu, \sigma)$, where $\mu = w_0 + w_{\text{height}} * x_{\text{height}} + w_{\text{weight}} * x_{\text{weight}} + w_{\text{age}} * x_{\text{age}}$. We placed a normal prior on the intercept centered on the observed average serve speed, and weakly informative $\text{Normal}(0, 5)$ priors were placed on each slope coefficient. A HalfNormal prior was also placed on the noise term σ , since it must be positive. The posterior was sampled running 4 independent chains with 2000 draws each and 1000 tuning steps. We assessed the convergence through $\hat{r}$ values (all 1.0) and trace plots, which confirmed reliable sampling. Instead of just a single point estimate, this model produces a full posterior distribution over serve speed and captures uncertainty. One limitation is that there is technically a nonzero chance of predicting a negative serve speed. However, it would require an input so many standard deviations away from the mean that it is practically impossible for any realistic player profile.

4. Data Collection

Our dataset consists of professional ATP players with their recorded average serve speed, height, weight, age, and dominant hand. The serve speed data was downloaded from Ultimate Tennis Statistics (ultimatetennisstatistics.com), and physical attributes were scraped from Tennis Explorer (tennisexplorer.com). These two sources were merged on player name to produce a single dataset. After removing players with missing physical measurements, our final dataset contained approximately 100 players. All physical attributes were standardized, and since all players in the dataset are male, gender was not included as a predictor.

5. ML Results and Interpretation

5.1 User-User Collaborative Filtering

After predicting on 10 randomly masked players, we found that a k value of 10 had the lowest RMSE and MAE for both cosine and L2 similarity scores. Cosine performed slightly better than L2 across all metrics. Since all RMSE and MAE values were greater than 0.1, we can say that our model was not very good at predicting serve speed based on the metrics available in the data. Of the three k -values, $k = 10$ performed the best. When the k value was 3, there probably was too much noise in the data which led to inaccurate predictions.

Further investigation was conducted after the model had been tested on the three k -values in order to find what the optimal k -value was by calculating the RMSE for a range of k -values between 1 to 20. The lowest amount of error was found at $k = 12$, after which the cosine RMSE began to increase. At $k = 12$, the RMSE was 0.148/0.149 (Cosine/L2), and the MAE was 0.117/0.118 (Cosine/L2). When the k -value was greater than 12, the RMSE for cosine spiked and showed a very odd result. At $k = 20$, the RMSE was approximately 1.2 which indicated a very high error and may have been a sign that the cosine similarity score had issues in its calculations.

	$k=3$	$k=5$	$k=10$
RMSE (Cosine)	0.1752709	0.1653842	0.1557466
RMSE (L2)	0.1774836	0.1655726	0.1576552
MAE (Cosine)	0.1464379	0.1362444	0.1261938
MAE (L2)	0.148791	0.1366654	0.1276063

Table 1: Cosine and L2 RMSE and MAE values for $k = 3, 5$, and 10

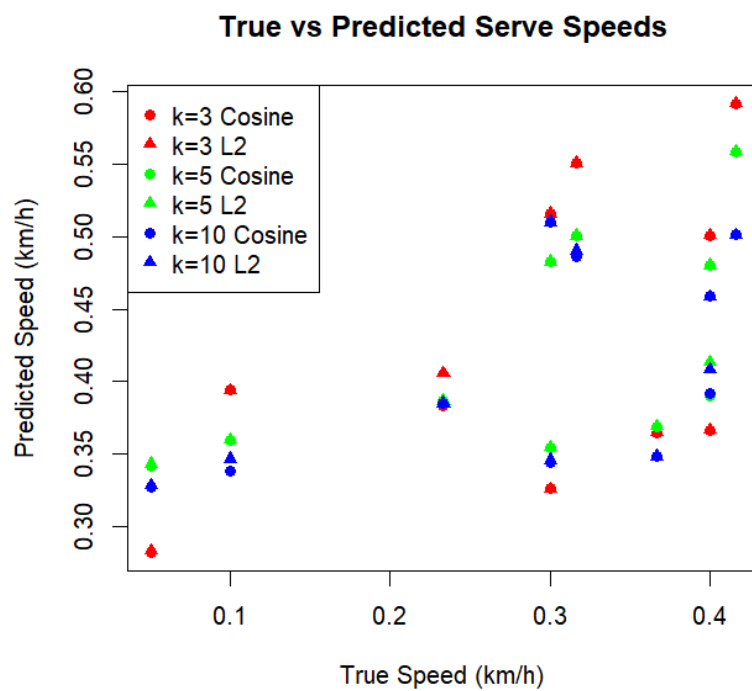


Figure 1: True vs Predicted Serve Speeds by User-User Collaborative Filtering

5.2 Bayesian Modeling

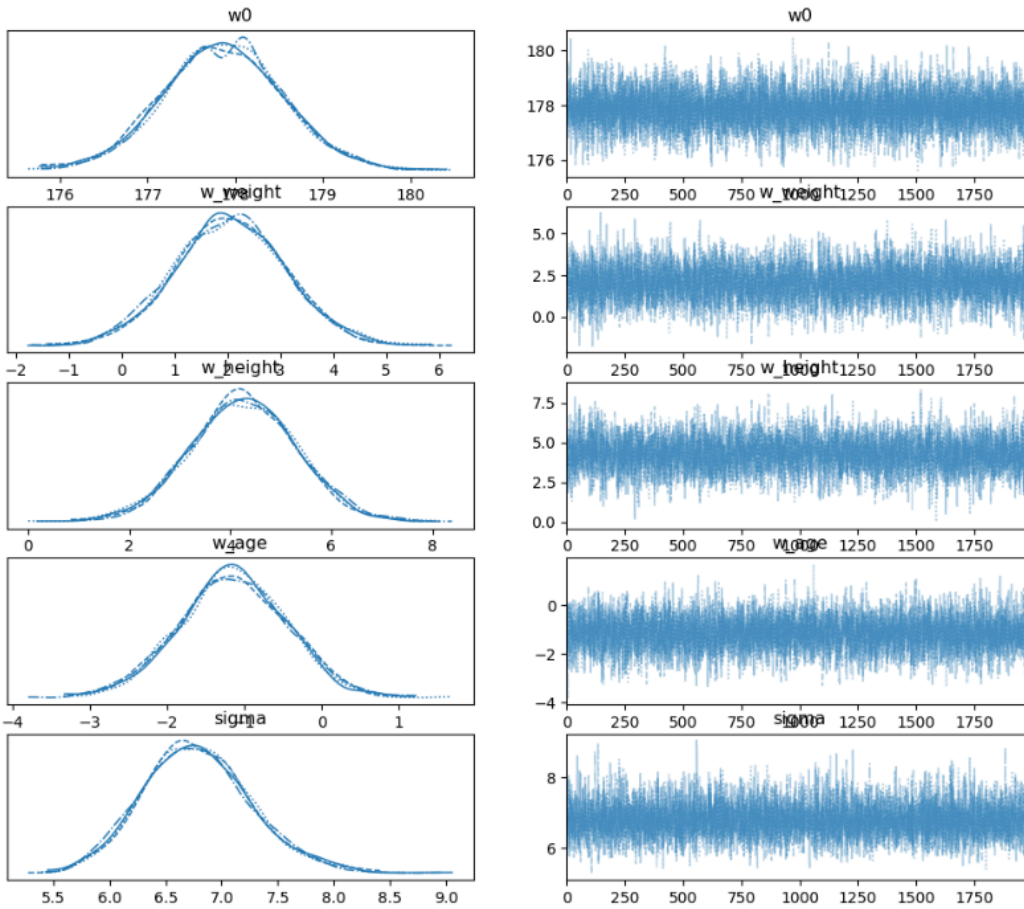


Figure 2: Posterior Distributions and Trace Plots

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
w0	177.86	0.68	176.64	179.23	0.01	0.01	10246.49	5277.37	1.0
w_weight	2.05	1.06	0.07	4.09	0.01	0.01	5406.21	5609.06	1.0
w_height	4.26	1.06	2.29	6.26	0.01	0.01	5249.43	5090.27	1.0
w_age	-1.15	0.70	-2.51	0.11	0.01	0.01	7498.68	5646.82	1.0
sigma	6.79	0.49	5.89	7.73	0.01	0.01	8412.00	6253.58	1.0

Table 2: Posterior Summary

The model converged successfully with r-hat values of 1.0 and the trace plots confirmed this. The intercept w_0 estimated a baseline serve speed of 177.9 km/h, which represents the

predicted speed for a player with average height, weight, and age. Height was the strongest positive predictor, where a one standard deviation increase corresponds to a +4.3 km/h change in serve speed. Weight showed a weaker positive effect, where a one standard deviation increase corresponds to a +2.1 km/h change in serve speed. Age had a small negative effect, where a one standard deviation increase corresponds to a -1.2 km/h change in serve speed. The residual noise term sigma was 6.8 km/h, indicating a meaningful amount of uncertainty in predictions.

6. Conclusions and Future Work

In order to make collaborative filtering a more useful model to predict a player's serve speed, it would be important to collect more data points and metrics. Because we only had about 100 players to base our predictions off of, it suffered from cold starts. The testing method also needed improvements, as the 10 random players were *all* masked before any predictions were made. This meant that there were 9 less data points that could be used to predict the target player. In future implementations of model testing, we should mask each player individually. It would also be good to iterate over the dataset multiple times with different selections of 10 random players, observing the variation in performance.

The Bayesian linear regression model showed that height and weight are meaningful positive predictors of serve speed, while age showed a small negative effect. However, the residual noise term of 6.8 km/h suggests that a portion of serve speed remains unexplained by just the physical attributes we used. This makes sense, as serve speed is a complex result of numerous other factors like mechanics, muscle composition, and technique. Given the small dataset size, this Bayesian approach was well suited for this problem since it naturally shows uncertainty in predictions. In the future, this model could be improved by incorporating additional metrics and modeling the effect of age nonlinearly, which would capture the natural peak of serve speed as a player ages.

Serves are not just dependent on a player's height and weight - body mechanics, strength, and racket metrics are important factors too. Our data only contained pro athletes, which makes the model assume that any user they are predicting for are professionals and not amateur tennis players. To improve our dataset, it would be important to collect data from all levels of tennis players. We can take more measurements such as arm length, racket speed, and various strength metrics such as bench press or grip strength. In order to observe body mechanics, a CNN method may be interesting to implement. An image could be captured at a particular point or points in the player's serve, and could then be associated with a particular serve speed so that the model can learn to predict speed from the images.

7. Appendix

7.1 Code

Github Repository

- <https://github.com/kenichigomi/DS4420-Final-Project>

R Implementation (Manual)

- User-User Collaborative Filtering
 - collab_filtering.R

```
user_collab_filter <- function(dataframe, target, similarity_metric, k) {  
  users_matrix <- as.matrix(dataframe)  
  tgt_idx <- target  
  
  if (target > nrow(users_matrix)) {  
    message(paste("Error:", target, "not found in the dataset."))  
    return(NULL)  
  }  
  
  if (!any(is.na(users_matrix[tgt_idx, ]))) {  
    message(paste(target, "is not missing any data"))  
    return(NULL)  
  }  
  
  row_means <- rowMeans(users_matrix, na.rm = TRUE)  
  data_centered <- users_matrix - row_means  
  
  data_centered[is.na(data_centered)] <- 0  
  
  if (!similarity_metric %in% c("L2", "cosine")) {  
    message("Invalid input for similarity metric. Please pick between 'L2' or 'cosine'.")  
    return(NULL)  
  }  
  
  } else if (similarity_metric == "L2") {  
    sim_list <- apply(data_centered, 1, function(row) {  
      sqrt(sum((row - data_centered[tgt_idx, ])^2))  
    })  
    sim_list[tgt_idx] <- NA  
  
    sim_list <- sim_list * -1
```



```

sim_scaled <- (sim_list - min(sim_list, na.rm = TRUE)) /
  (max(sim_list, na.rm = TRUE) - min(sim_list, na.rm = TRUE))

} else if (similarity_metric == "cosine") {
  tgt_vec <- data_centered[tgt_idx, ]
  sim_list <- apply(data_centered, 1, function(row) {
    sum(tgt_vec * row) / (sqrt(sum(tgt_vec^2)) * sqrt(sum(row^2)))
  })
  sim_list[tgt_idx] <- NA

  sim_scaled <- (sim_list - min(sim_list, na.rm = TRUE)) /
    (max(sim_list, na.rm = TRUE) - min(sim_list, na.rm = TRUE))
}

k_similar_idx <- order(sim_scaled, decreasing = TRUE, na.last = NA)[1:k]
k_similar_scores <- sim_scaled[k_similar_idx]

na_positions <- which(is.na(users_matrix[tgt_idx, ]))

for (col in na_positions) {
  num <- 0
  denom <- 0

  for (i in seq_along(k_similar_idx)) {
    similar <- k_similar_idx[i]
    score <- k_similar_scores[i]

    if (!is.na(users_matrix[similar, col])) {
      num <- num + users_matrix[similar, col] * score
      denom <- denom + score
    } else {
      num <- num + mean(users_matrix[similar, ], na.rm = TRUE) * score
      denom <- denom + score
    }
  }
}

users_matrix[tgt_idx, col] <- num / denom
}

return(users_matrix[tgt_idx, ])
}

```

Python Implementation

- Bayesian Modeling
 - bayesian_ml.ipynb

with pm.Model() as model:

```
w0 = pm.Normal("w0", mu=mean_speed, sigma=std_speed) # intercept ~ average serve speed
w_weight = pm.Normal("w_weight", mu=0, sigma=5)
w_height = pm.Normal("w_height", mu=0, sigma=5)
w_age = pm.Normal("w_age", mu=0, sigma=5)
sigma = pm.HalfNormal("sigma", sigma=10)

mu = (
    w0 +
    w_weight * X_weight +
    w_height * X_height +
    w_age * X_age
)

speed_obs = pm.Normal("speed_obs", mu=mu, sigma=sigma, observed=tennis["speed_kmh"])

trace = pm.sample(draws=2000, tune=1000, chains=4, target_accept=0.9, return_inferencedata=True)
```

7.2 Bibliography

- Hernández-Davó, Jose Luis, Francisco Javier Moreno, David Sanz-Rivas, Héctor Hernández-Davó, Álvaro Coves, and Carla Caballero. 2019. "Variations in Kinematic Variables and Performance in the Tennis Serve According to Age and Skill Level." *International Journal of Performance Analysis in Sport* 19 (5): 749–62. doi:10.1080/24748668.2019.1653036.
- Vaverka, Frantisek, and Miroslav Cernosek. 2013. "Association between Body Height and Serve Speed in Elite Tennis Players." *Sports Biomechanics* 12 (1): 30–37. doi:10.1080/14763141.2012.670664.
- Wong, F. K. H., J. H. Keung, N. M. Lau, D. K. Ng, J. W. Chung, and D. H. Chow. "Effects of Body Mass Index and Full Body Kinematics on Tennis Serve Speed." *Journal of Human Kinetics* 40 (2014): 21–28. doi:10.2478/hukin-2014-0003